

MelodyHub

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Project Description

In this project, we plan to build a music website app. Specifically, we want to include the following features:

1. Registration and Login feature
 - a. Website built-in registration and login feature.
 - b. Registration and login feature using Google.
2. Music Play
 - a. Play the track the user selects.
 - b. The user can access the details of each song. When the user is browsing other information on the website, the play bar will be minimized to the foot of the website.
 - c. (Optional, if time permits) The lyrics of the track being played will be displayed and automatically scrolled.
3. Playlist
 - a. Users can create playlists, name their playlists, and add/delete tracks to/from the playlists.
 - b. "My Favorites" is a special playlist. For each song, there is an "add to favorites" button that users can add/delete tracks to/from their favorites.
 - c. "Recent" is also a special playlist that will keep track of the recent 100 songs the users played.
 - d. Inside each playlist, users can sort tracks by default (the time the track was added to the playlist), the track name, and the artist name.
4. Search
 - a. Users can search tracks and artists based on their names
 - b. (Optional, if time permits) Users can search for other users based on their usernames and follow/unfollow.
5. Listen Together (optional, if time permits)
 - a. Users can create 'Listen Together' rooms and invite others to join. The creator of the room selects a song to play, and all joined users will listen to the track in real-time synchronization, ensuring everyone is at the same point in the song.
6. User and Admin account
 - a. User account: User account has the features 1 to 5 listed above.
 - b. Admin account: Admin account has the feature to add(upload) new tracks (track name, track audio, track artist, track lyric (optional), album cover) to the database backend. Admin accounts can suspend or reactivate user accounts.

Technology

- Django is selected for its comprehensive framework that supports rapid development, offering out-of-the-box solutions for user authentication and content management.
- OAuth supports secure Google Sign-Up and Sign-In by allowing users to authenticate with their Google accounts, ensuring a streamlined and secure login process.
- WebSockets provide real-time communication capabilities, essential for live song playback and instant updates to playlists and favorites.
- AJAX enables asynchronous web page updates, which is critical for a responsive user interface that allows users to interact with MelodyHub without constant page reloads.
- jQuery simplifies client-side scripting, optimizing the platform's interactivity and responsiveness. It aids MelodyHub in dynamically updating content.